

1. Foundation: HTML, CSS, Responsive Design

Goal: Create a simple responsive landing page (e.g., a “Ceylinco Life” style info page).

YouTube Resources:

- **freeCodeCamp** – “Responsive Web Design – Full Course” (4 hours)
→ Covers HTML5, CSS3, Flexbox, Grid, media queries.
- **Traversy Media** – “Build An HTML5 Website With A Responsive Layout” (1 hour)
→ Practical project you can code along with.

GitHub:

Create a repo named `ceylinco-landing-page` and upload your finished page.

2. CSS Frameworks: Tailwind CSS & Bootstrap

Goal: Convert your landing page to use Tailwind CSS, then rebuild a small component using Bootstrap to understand the difference.

YouTube Resources:

- **Tailwind CSS** – Official playlist: “Tailwind CSS From Scratch” (by Traversy Media)
- **Bootstrap** – “Bootstrap 5 Crash Course” (by Traversy Media)

GitHub:

Create a new repo `tailwind-bootstrap-practice` and include both versions.

3. JavaScript & Modern Front-end Tooling

Goal: Build a simple interactive widget (e.g., a premium calculator) using vanilla JS, then move to React.

YouTube Resources:

- **The Net Ninja** – “Modern JavaScript (ES6) Tutorial” (playlist)
- **freeCodeCamp** – “React JS Crash Course 2024” (1.5 hours)
- **Next.js** – “Next.js 15 Tutorial” by CodeWithAntonio or “Next.js for Beginners” by freeCodeCamp

GitHub:

- Repo 1: `vanilla-js-premium-calculator`
 - Repo 2: `nextjs-demo-app` (e.g., a simple blog or policy list)
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4. Back-end: Node.js & Express.js

Goal: Build a REST API that can handle GET and POST requests (e.g., a “contact” endpoint that stores messages).

YouTube Resources:

- **Traversy Media** – “Node.js / Express Crash Course” (1.5 hours)
- **Academind** – “Node.js – The Complete Guide” (first few sections are free)

GitHub:

Repo: `node-express-api` with simple endpoints.

5. Databases: MySQL, MSSQL, MongoDB

Goal: Learn SQL basics with MySQL (or a free online SQL sandbox). Then add a MongoDB database to your Node.js API.

YouTube Resources:

- **MySQL** – *"MySQL Tutorial for Beginners"* by Programming with Mosh
- **MongoDB** – *"MongoDB Crash Course"* by Traversy Media
- **MSSQL** – Since MSSQL is used at Ceylinco Life, watch *"SQL Server Tutorial"* by freeCodeCamp (covers T-SQL basics).

GitHub:

Update your Node.js API to connect to MongoDB and MySQL (separate branches). Document the setup in the README.

6. Full-Stack MERN Project

Goal: Combine everything into a MERN (MongoDB, Express, React, Node.js) application. A simple "Policy Management" demo where you can add, view, and delete policy records.

YouTube Resources:

- **freeCodeCamp** – *"MERN Stack Course 2024 – Build a Real Project"* (4 hours)
- **The Net Ninja** – *"MERN Stack Playlist"*

GitHub:

Repo: `mern-policy-manager` – include both front-end and back-end folders, with clear instructions.

7. Generative AI (Bonus)

Goal: Show interest by adding a simple AI feature—e.g., use OpenAI's API to generate a summary or a greeting message.

YouTube Resources:

- **OpenAI** – *"Getting Started with OpenAI API"* (official channel)
- **Web Dev Simplified** – *"How to Use OpenAI API in Node.js"*

GitHub:

Add a small "AI-assisted" endpoint to your Node.js app. Even a simple call to GPT-3.5 that returns a response shows initiative.

How to Structure Your GitHub Portfolio

- Create a **single GitHub profile** with all the repositories.
 - Write a **good README** for each project:
 - What the project does
 - Technologies used
 - How to run it locally
 - Screenshot (if front-end)
 - Pin the best 3-6 repos to your profile.
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Learning Tips

1. **Code along** with the videos – don't just watch.
2. **Commit often** – after each small feature, push to GitHub.
3. **Ask questions** – use YouTube comments, Stack Overflow, or communities like r/learnprogramming.
4. **Don't aim for perfection** – focus on getting a working project that shows you can learn and build.